

 Quick Recall (Ch. 10.1–10.2)

The counting method (10.1). We have a set of objects, and we want to show that there exists some “good” object among them (with some required property). We count how many objects there are in total, and we upper-bound the number of bad objects. If the number of bad objects is smaller than the number of all objects, then some good object must exist.

Example (Dovetail shuffles, 10.1.1). There are $52!$ orderings of a deck. One round of dovetail shuffling produces at most $\binom{52}{26}^4$ orderings; so 4 rounds yield at most $\binom{52}{26}^4 < 52!$. Hence some ordering of the deck cannot be obtained by 4 shuffles.

Stirling (for comparisons with $n!$). $n! \sim \sqrt{2\pi n} (n/e)^n$. In particular, $n!/C^n \rightarrow \infty$ for any fixed constant C .

Definition 10.2.2 (the space C_n). The probability space C_n has as elementary events all n -term 0/1 sequences (i.e., elements of $\{0,1\}^n$), all with equal probability $1/2^n$. It models n coin tosses.

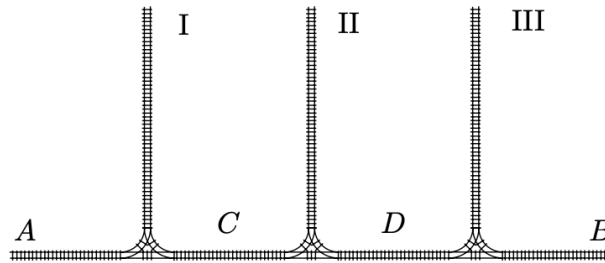
Definition 10.2.6 (independence). Events $A_1, A_2, \dots, A_n \subseteq \Omega$ are called *independent* if for every $I \subseteq \{1, 2, \dots, n\}$,

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i).$$

The A_i in C_n . Define $A_i \subseteq \{0,1\}^n$ as the set of all sequences with 1 in the i -th position. The textbook states (without proof) that $A_1, A_2, \dots, A_n$ are independent.

 Practice: Textbook Exercise 10.1.4* — the train cars

*In the following diagram, a train with n cars is standing on the rail track labeled A . The cars are being moved to the track B . Each car may or may not be shifted to some of the side tracks I–III but it should visit each side track at most once and it should go through tracks C and D only once.



Prove that if n is large enough then there is some order of the cars that cannot be achieved on track B .

 Hints

Hint (textbook). Code all possible ways of the cars passing through the depicted system of rails, e.g. “a car from rail A goes to rail I , car from A to C , car from I to C , car from C to II , ...”. Show that at most C^n possible codes exist (for a large enough constant C); hence there are at most this many possible car orders on rail B , and this is smaller than $n!$ for n large.

Suggested steps. (1) Each car makes at most a bounded number L of moves (constant in n , since it visits I, II, III at most once and C, D at most once); so the total code length is at most Ln . (2) At each step the move-type has at most k choices (a constant determined by the rail topology). Pad each code with a “null” symbol to uniform length Ln ; this gives $\#\{\text{codes}\} \leq (k+1)^{Ln} =: C^n$. (3) $\#\{\text{orderings on } B\} \leq \#\{\text{codes}\} \leq C^n$. Compare with $n!$ using $n!/C^n \rightarrow \infty$ (Stirling).

Answer: Step 1: Total code length is at most Ln . Each car begins on A , ends on B , visits each of I, II, III at most once and each of C, D at most once. So it visits at most 7 tracks and makes at most $L := 6$ track-to-track moves – a bound independent of n . A *code* lists, in temporal order, every move made by every car (as in the hint). Since each of the n cars contributes at most L moves, the code length is at most Ln .

Step 2: $\#\{\text{codes}\} \leq C^n$. At each step, the move-type ($X \rightarrow Y$) comes from a fixed finite set $\mathcal{T}$ of allowed track-to-track transitions – determined by rail topology, not by n . Let $k := |\mathcal{T}|$. Pad each code with a “null” symbol (a do-nothing step) until its length is exactly Ln ; each of the Ln positions then carries one of $k+1$ symbols, and distinct codes pad to distinct padded codes. Hence

$$\#\{\text{codes}\} \leq (k+1)^{Ln} = ((k+1)^L)^n =: C^n,$$

a constant power – the bound holds for every $n \geq 1$.

Step 3: Compare with $n!$. Every ordering of cars on B is realized by at least one code, so $\#\{\text{orderings}\} \leq \#\{\text{codes}\} \leq C^n$. By Stirling, $n!/C^n \sim \sqrt{2\pi n} (n/(eC))^n \rightarrow \infty$, so $n! > C^n$ for n large enough – and some ordering of cars on B cannot be realized. $\square$

Remark (teaching tip). Same counting method as Example 10.1.1 (dovetail shuffles): count $\#_{\text{total}}$ ($n!$) and upper-bound $\#_{\text{achievable}} (\leq C^n)$. The crude bound C^n ignores routing constraints but suffices since $n!$ beats any exponential.

- (a) Show that the events $A_1, A_2, \dots, A_n$ in the probability space C_n defined in the text following Definition 10.2.6 are really independent.
- (b)* Let (Ω, P) be a finite probability space. Suppose that n independent events $A_1, A_2, \dots, A_n \subseteq \Omega$ exist such that $0 < P(A_i) < 1$ for each i . Show that then $|\Omega| \geq 2^n$.

Hints

Hint (textbook, for (b)). If we replace some of the A_i by their complements we still have a collection of independent events, and hence the intersection of all these events has a nonzero probability. In this way, one can produce 2^n disjoint events with nonzero probability.

Suggested steps for (a). (1) Compute $P(A_i)$ and $P(\bigcap_{i \in I} A_i)$ by counting sequences. (2) Check the product condition matches for every I .

Suggested steps for (b). (1) For each $\varepsilon \in \{0, 1\}^n$, define B_ε as the intersection: A_i if $\varepsilon_i = 1$, else $\Omega \setminus A_i$. (2) Generalize HW 10.2.6 to n events (same calculation as HW 10.2.6) to argue $P(B_\varepsilon) > 0$. (3) Show pairwise disjointness via any differing coordinate, then conclude $|\Omega| \geq 2^n$.

Answer: Part (a). By Definition 10.2.2, the elementary events of C_n are sequences in $\{0, 1\}^n$, each of probability $1/2^n$. So for any event $E \subseteq \{0, 1\}^n$, $P(E) = |E|/2^n$.

Step 1: $P(A_i) = 1/2$. The set A_i consists of sequences with the i -th position fixed to 1 and the other $n - 1$ positions free, so $|A_i| = 2^{n-1}$ and $P(A_i) = 2^{n-1}/2^n = 1/2$.

Step 2: $P(\bigcap_{i \in I} A_i) = 2^{-|I|}$. For any $I \subseteq \{1, \dots, n\}$, $\bigcap_{i \in I} A_i$ consists of sequences with 1 in every position of I ; the remaining $n - |I|$ positions are free. So

$$P\left(\bigcap_{i \in I} A_i\right) = \frac{2^{n-|I|}}{2^n} = 2^{-|I|}.$$

Step 3: Match. Also $\prod_{i \in I} P(A_i) = (1/2)^{|I|} = 2^{-|I|}$. So $P(\bigcap_{i \in I} A_i) = \prod_{i \in I} P(A_i)$ for every I , and by Definition 10.2.6 the events $A_1, \dots, A_n$ are independent. $\square$

Part (b).

Step 1: Define the cells. For each $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{0, 1\}^n$, set

$$B_\varepsilon := \bigcap_{i: \varepsilon_i=1} A_i \cap \bigcap_{i: \varepsilon_i=0} (\Omega \setminus A_i).$$

Step 2: $P(B_\varepsilon) > 0$ for every ε . We claim: replacing any A_i with $\Omega \setminus A_i$ in a mutually independent collection preserves mutual independence. This generalizes HW 10.2.6 via the same one-line calculation: for any $J \subseteq \{1, \dots, n\} \setminus \{i\}$,

$$P((\Omega \setminus A_i) \cap \bigcap_{j \in J} A_j) = P(\bigcap_{j \in J} A_j) - P(A_i \cap \bigcap_{j \in J} A_j) = (1 - P(A_i)) \prod_{j \in J} P(A_j),$$

using independence on J and on $\{i\} \cup J$; this is the required product condition involving $\Omega \setminus A_i$ (and for index sets not containing i , the condition is inherited unchanged from the original). Iterating for each i with $\varepsilon_i = 0$, the events $A_i^{(\varepsilon)} := A_i$ (if $\varepsilon_i = 1$) or $\Omega \setminus A_i$ (if $\varepsilon_i = 0$) are mutually independent. Hence

$$P(B_\varepsilon) = \prod_{i=1}^n P(A_i^{(\varepsilon)}) = \prod_{i: \varepsilon_i=1} P(A_i) \cdot \prod_{i: \varepsilon_i=0} (1 - P(A_i)) > 0,$$

since $0 < P(A_i) < 1$ makes each factor strictly positive.

Step 3: $B_\varepsilon \cap B_{\varepsilon'} = \emptyset$ when $\varepsilon \neq \varepsilon'$. If $\varepsilon \neq \varepsilon'$, they differ at some coordinate j . WLOG $\varepsilon_j = 1$ and $\varepsilon'_j = 0$. Then $B_\varepsilon \subseteq A_j$ and $B_{\varepsilon'} \subseteq \Omega \setminus A_j$, and these are disjoint.

Step 4: Conclude $|\Omega| \geq 2^n$. Each B_ε has $P(B_\varepsilon) > 0$, so $B_\varepsilon \neq \emptyset$ – it contains at least one elementary event. The 2^n sets $\{B_\varepsilon : \varepsilon \in \{0, 1\}^n\}$ are pairwise disjoint, so Ω contains at least 2^n distinct elementary events. Therefore $|\Omega| \geq 2^n$. $\square$

Remark (teaching tip). The bound is tight: C_n itself attains $|\Omega| = 2^n$ with n nontrivial independent events (by part (a)). So 10.2.8 characterizes C_n as the smallest probability space supporting n nontrivial independent events.